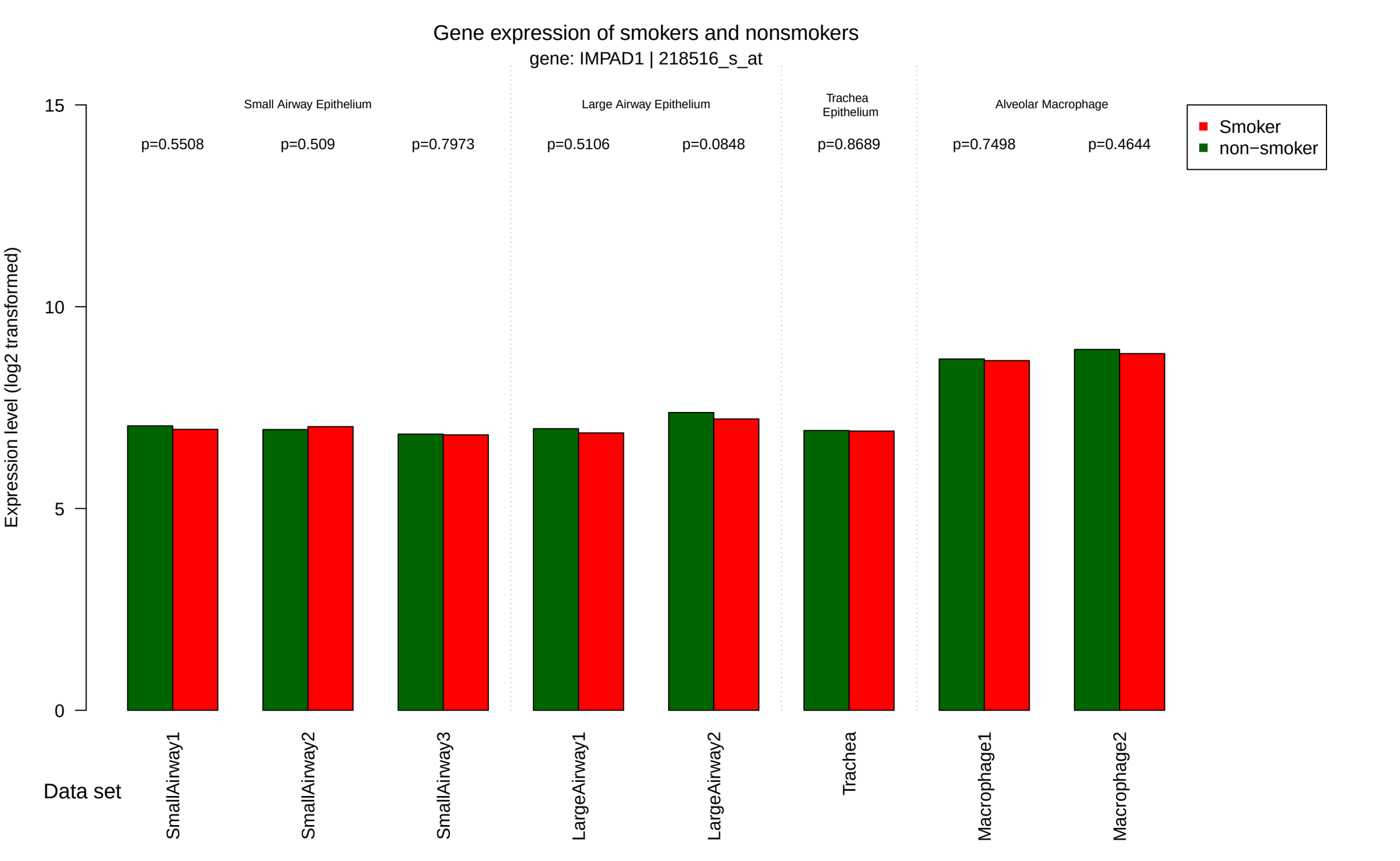
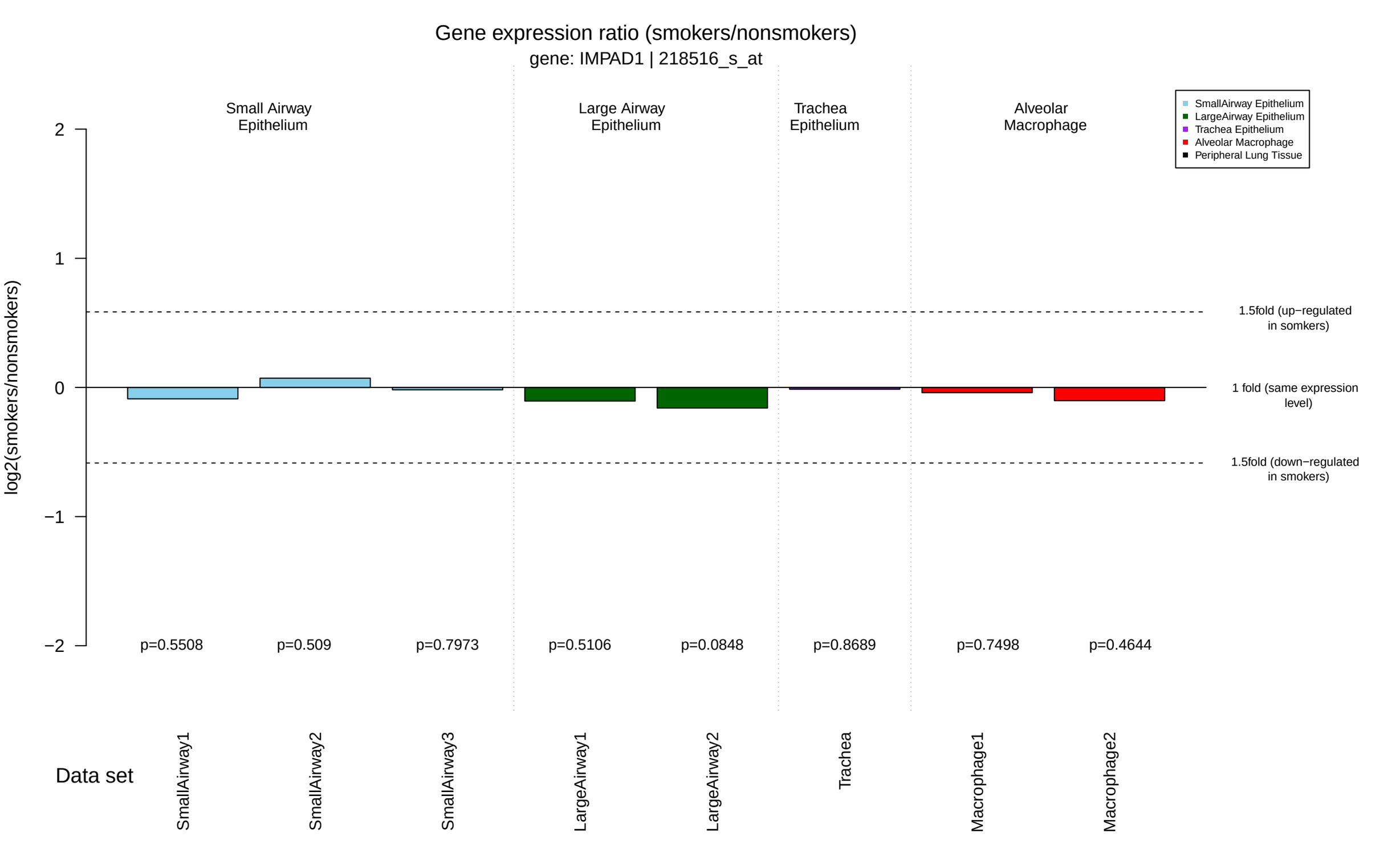
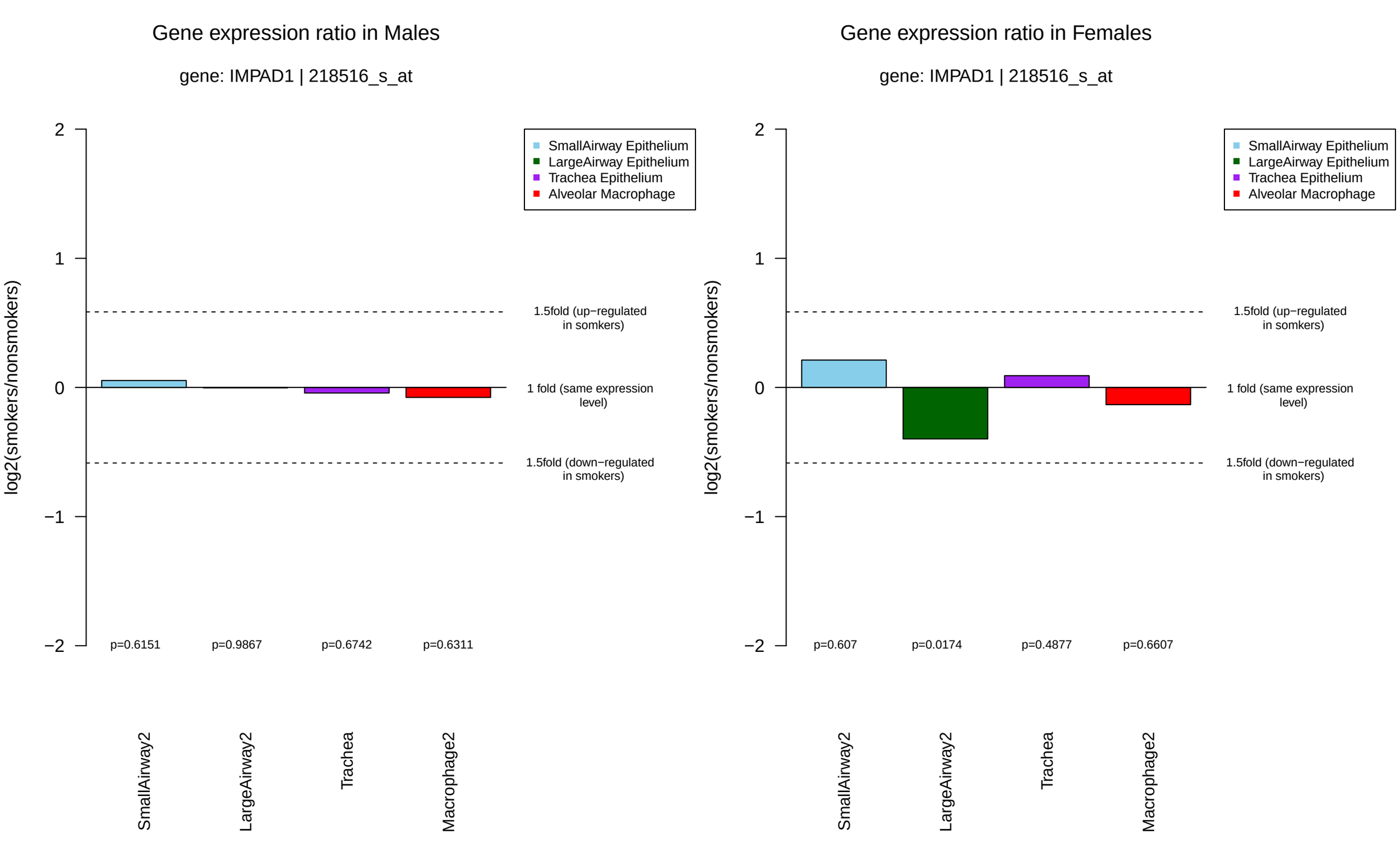
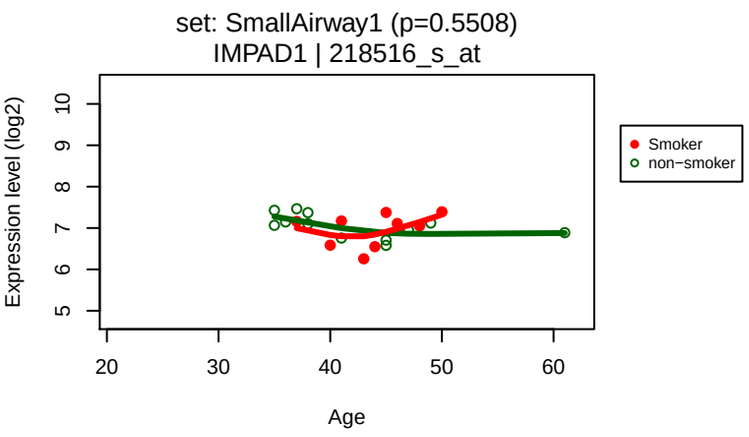
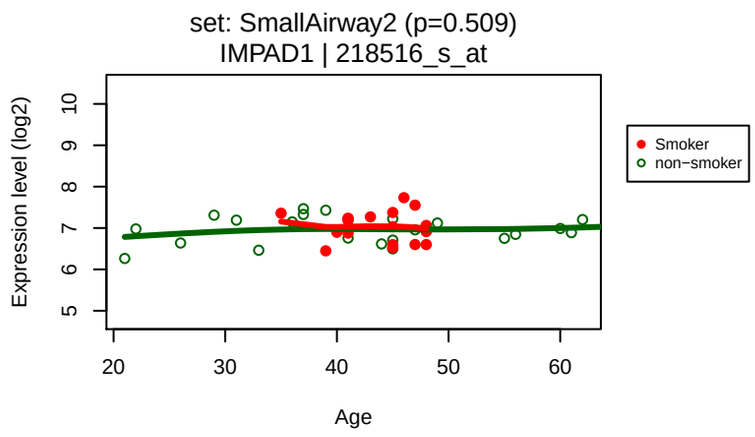
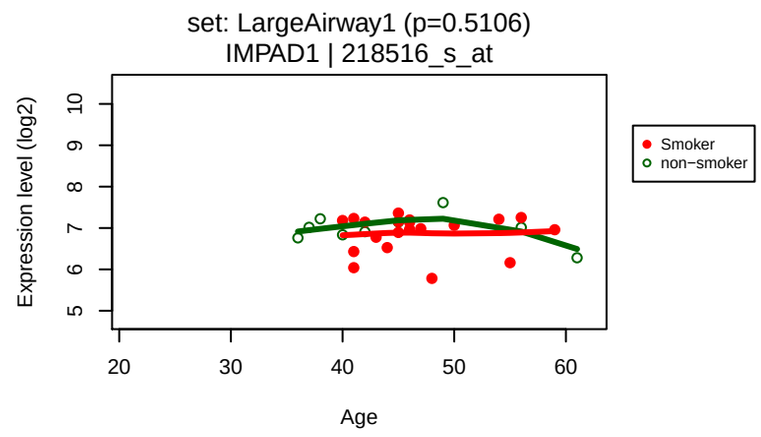

Gene expression of smokers and nonsmokers
gene: IMPAD1 | 218516_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5508
p=0.509
p=0.7973
p=0.5106
p=0.0848
p=0.8689
p=0.7498
p=0.4644
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

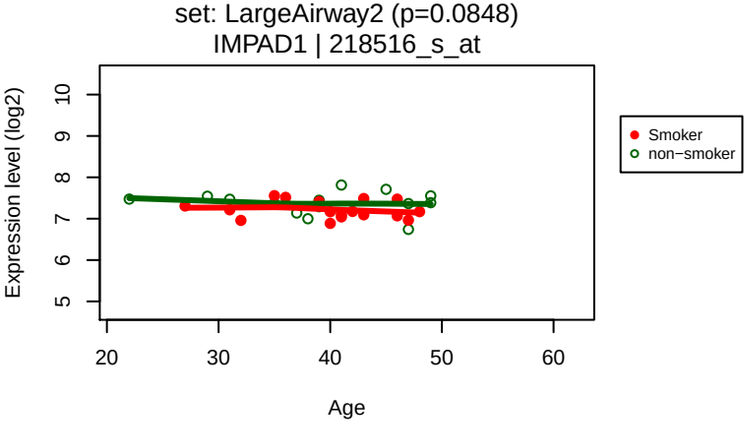

Gene expression ratio (smokers/nonsmokers)
gene: IMPAD1 | 218516_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5508
p=0.509
p=0.7973
p=0.5106
p=0.0848
p=0.8689
p=0.7498
p=0.4644
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

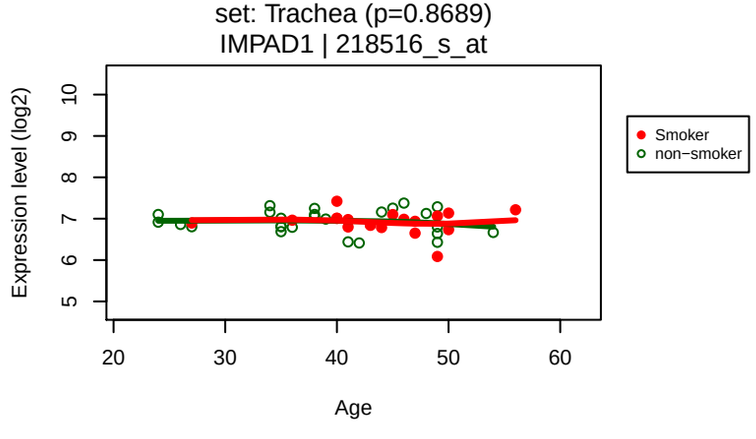

Gene expression ratio in Males
gene: IMPAD1 | 218516_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6151
p=0.9867
p=0.6742
p=0.6311
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IMPAD1 | 218516_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.607
p=0.0174
p=0.4877
p=0.6607
SmallAirway2
LargeAirway2
Trachea
Macrophage2

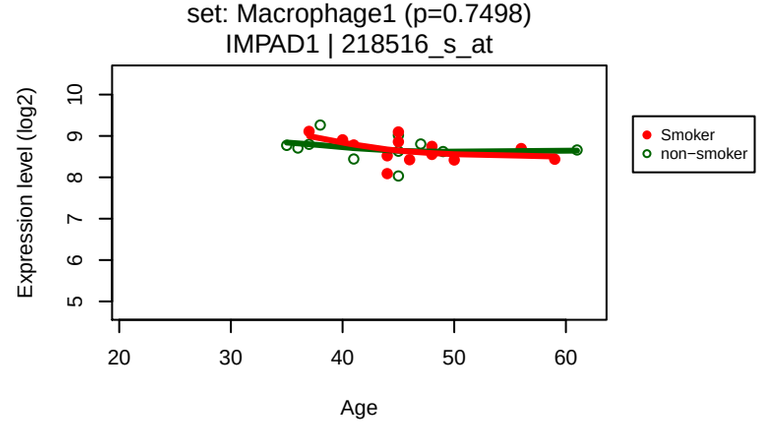

set: SmallAirway1 (p=0.5508)
IMPAD1 | 218516_s_at
Expression level (log2)
Age
Smoker
non−smoker

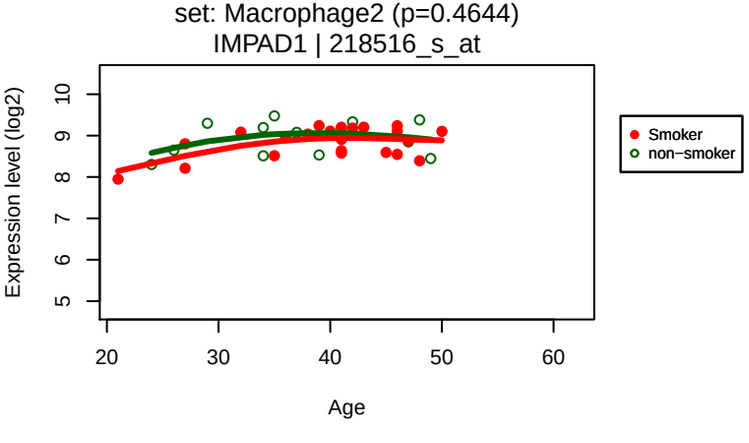

set: SmallAirway2 (p=0.509)
IMPAD1 | 218516_s_at
Expression level (log2)
Age
Smoker
non−smoker

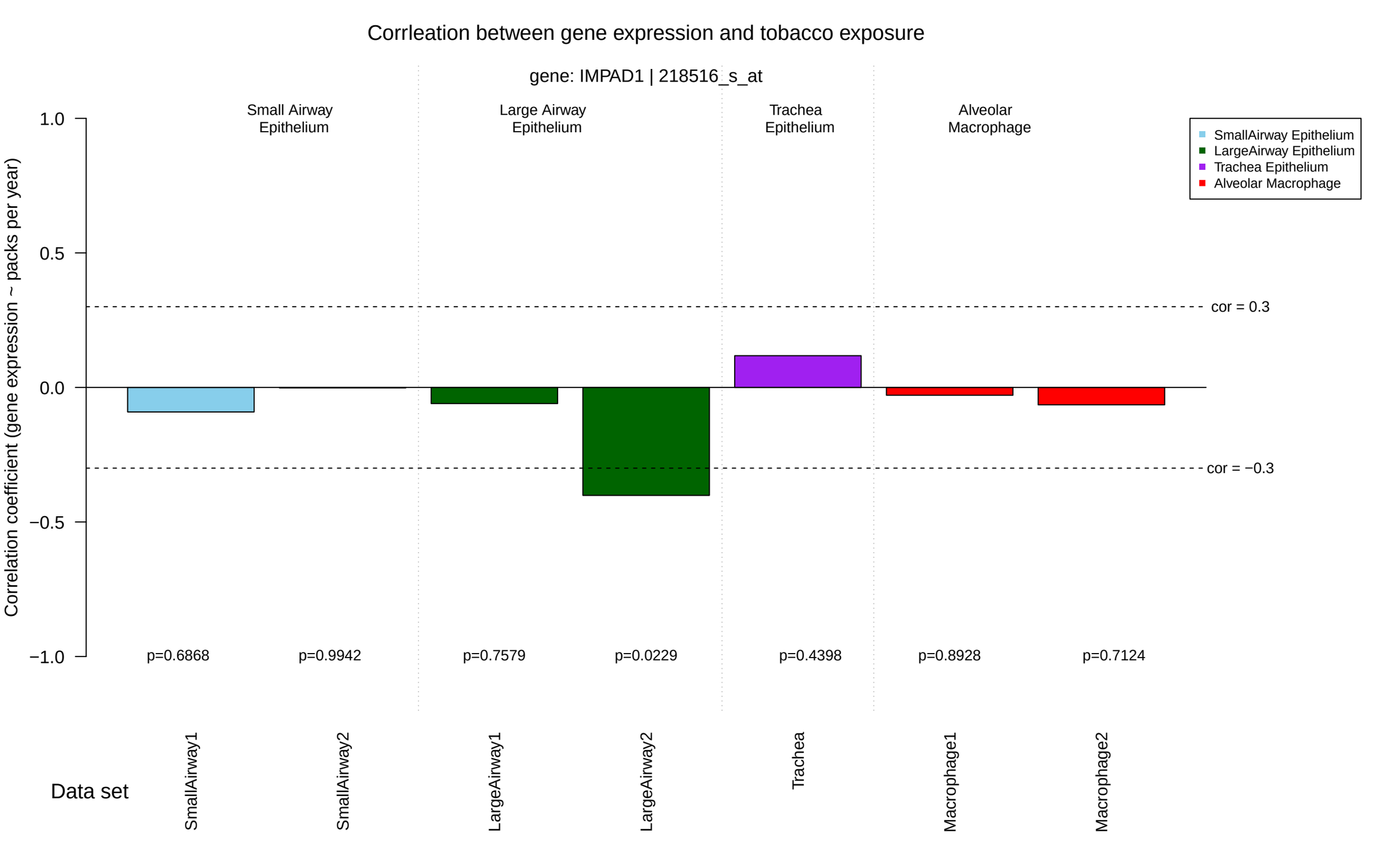

set: LargeAirway1 (p=0.5106)
IMPAD1 | 218516_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0848)
IMPAD1 | 218516_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8689)
IMPAD1 | 218516_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7498)
IMPAD1 | 218516_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4644)
IMPAD1 | 218516_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IMPAD1 | 218516_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6868
p=0.9942
p=0.7579
p=0.0229
p=0.4398
p=0.8928
p=0.7124
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2